Proceedings of the 9th International Conference on Intelligent Computing and Control Systems (ICICCS-2026)
DVD Part Number: CFP26K74-DVD; ISBN: 979-8-3315-8946-2

Design and Implementation of An IoT-Enabled Flood Detection and Notification System

R. Panneerselvi
Department of CSE
Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology, Avadi,
Chennai-600062, India
panneerselvir@veltech.edu.in

Varshinii S. N
Department of CSE
Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology, Avadi,
Chennai-600062, India
varshiniisn06@gmail.com

Zebaanjum N
Department of CSE
Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology, Avadi,
Chennai-600062, India
zebaa3497@gmail.com

N.R.Rajalakshmi
Department of CSE
Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology, Avadi,
Chennai-600062, India
rajirajasekaran@gmail.com

Sumithra.M
Department of CSE
Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology, Avadi,
Chennai-600062, India
sumithra@veltech.edu.in

Seethalakshmi.K
Department of CSE
Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology, Avadi,
Chennai-600062, India
seethalakshmik@veltech.edu.in

***Abstract*—The increased risk of flooding in certain areas makes it necessary to use flood monitoring and alerting systems, especially during December and near major water bodies. This paper provides a real-time flood alerting system using IoT to give exact accuracy and a measurement of water level data for early flood detection. The developed system always provides an effective solution for flood warnings, helping the people who are in danger. Despite the advantages, implementing the system comes with challenges like high costs and regular maintenance. Using the Wi-Fi module, the system will send notifications through the Telegram app if the water level exceeds a particular level. The proposed system was functionally validated in a real-time setup by performing a water level detection under various circumstances.**

Keywords: IoT, Flood Alerting System, Arduino, Ultrasonic Sensor, Wi-Fi Module, Real-Time Monitoring, Telegram Notification, Disaster Management, Early Warning System, Environmental Monitoring.

I. Introduction

The IoT-Based Flood Alerting System is designed as an IoT Enabled Flood Detection and Notification System, which is based on Internet of Things(IoT) technology and uses Arduino and a Wi-Fi module. In India, monsoon rainfall occurs from June to October. Mostly in Chennai, heavy rainfall occurs between October and December.

Flooding causes severe environmental damage, including loss of human life, infrastructure damage and disruption of natural ecosystems. Geographically, it will affect the type of soil, environmental changes, modification of the ecosystem, and precipitation can play a part in flooding. In 2018, Kerala was declared a disaster-affected region after it was hit by massive monsoon rains.

The loss of human life and damage to the natural ecosystem are devastating consequences of flooding. This leads to infrastructure damage and human factors. Factors such as the volume and speed of the rain, geology, and moisture levels play a major role in determining the flood.

Due to the increased frequency and intensity of floods, climate change leads to loss of life, property, and infrastructure. To overcome this issue, flood alerting systems are required. The blending of an IoT with a flood monitoring system leads to an innovation or development idea for enabling real-time monitoring and data collection from various parameters. The combination of wireless sensor networks with IoT technology directs the transmission of data to a server for evaluation, and the alerts will be sent to the people who are at risk.

Direct Human Monitoring by Local Authorities, Municipality Workers, or community workers will often monitor the water level in rivers and drainage systems. Upon detecting a risk of flooding, higher authorities are informed to take precautions early and initiate the alerting process. Government or Meteorological Departments utilise satellite images, sensors, and weather models to detect or predict the rainfall and declare the warnings. These warnings are communicated to the people through the radio, television, and some mobile SMS messages.

Ultrasonic Sensors detect the water levels and send the information to the Arduino. Then Arduino will check whether the water level is in the correct stage or has reached the threshold value. If the water level reaches the threshold value, an Arduino will send the data to the Wi-Fi module. Then Wi-fi module send the alert messages to the people. Simultaneously the buzzer will make a sound to alert the people, and the LED glows red to indicate danger.

The objective of the proposed work is all about to design an IoT-based flood alerting system and to detect flood risk using threshold values with timely alerts through visual,auddible and mobile-based notifications.

The primary challenge addressed in this research leads to the development of a Flood Alerting System with low-cost, real time monitors and alerts. Based on the limitations identified in the existing system, this work focuses on real-time monitoring, low cost and simplicity. By installing the units in each area leads to monitoring the water level and sends alert notifications to the particular area by offering particular localised accuracy.

II. Literature review

In the development of this project, several IEEE papers were reviewed, and the key data points are briefly explained below:

The paper Flood and Landslide Prediction presents a disaster management system integrated with machine learning, which is used to give real-time alerts.A Flask-Web application and GSM module are used to enable the SMS notification by giving alerts and audio alarms. Additionally, a CNN model is used to review the ground-level images to check the water level and to detect risks like water overflow.

Recently, an IoT-Based Flood Monitoring System was developed in Thailand. The system includes a wireless network to measure the level of water and send the data using GPRS to the server, which gives real-time data. The study concludes that modern sensors can track the real-time water level to send alerts to the people.

Another study presents an IoT-Based Automated Flood Warning System using a Microcontroller and ESP8266. The system focuses on checking the water level, temperature, and humidity to determine whether a flood will occur or not. Sensors measure the water level, and the data is handled by a Microcontroller. When the water reaches a particular value, the alerts are indicated through an LED that glows red and a buzzer makes alarm sounds. This prototype uses the ThingSpeak IoT platform to send real-time alerts and send warnings to a particular area.

III. Related work

Many flood alerting system rely on devices such as radar, satellites, and sensors, and these devices are overpriced and sophisticated. The latest improvement in IoT technology offers a good option with good potential and substitutes. Previous research has demonstrated the application of IoT for environmental monitoring, but few studies have applied it to flood alerting systems specifically. This study expands on earlier research by blending IoT principles with Arduino to create a cost-effective and scalable solution.

A study by Al-Fuqaha et al.(2018) introduced a flood detection system using a wireless sensor network (WSNs) to provide real-time updates. The system uses an Ultrasonic sensor to measure the water level, and the data will be processed by the Arduino as a Microcontroller. Data from the Ultrasonic Sensors was sent to the communication devices like GSM and Wi-Fi without wires. The system successfully explained that it will alert the people and authorities when the water reaches beyond a particular level through mobile devices, which helps to reduce disaster consequences.

Future Developments in Technology: The integration of advanced IoT technology with the Flood Alerting System enables the measurement of water level, soil moisture, and pressure. Integrating the Machine Learning algorithms with the Flood Alerting System can improve the accuracy by using the historical patterns and environmental data to accurately predict data.

The proposed system depends on fixed threshold-based water level detection to ensure simplicity, low cost and real time response. Predictive flood models need data sets and machine-learning algorithms, which leads to an increase in system cost and complexity. Therefore, the present system depends on real-time detection and immediate alert. The combination of predictive and machine-learning algorithms is considered a future enhancement.

IV. System Design

To address the challenges, the architecture of the proposed system integrates with the IoT technology and the hardware components to provide real-time flood alerts through a buzzer and a display in an I2C circuit.

A. Architecture overview

As shown in Figure 1, the system framework consists of water-level sensors, an Arduino microcontroller, and a Wi-Fi module. The water-level sensors measure the water levels, and the Arduino receives the water level data to analyse the risk of flooding. The Wi-Fi module sends the message to the authorities and people at risk.

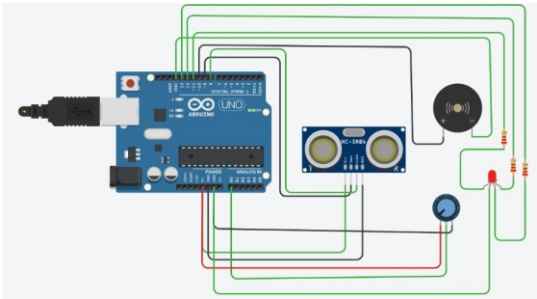

Fig. 1. Architecture Diagram

B. Hardware Components

The flood alerting system consists of several hardware devices, including a Microcontroller, Water-level sensors, Water pressure sensors, a Wi-Fi module, and components required for power supply. Some key features include:

- Real-time flood detection.
- Automated Data sharing.
- Accurate sensor readings.
- Localised Accuracy.

Key functions of Hardware components:

1) Arduino Microcontroller : Collects readings from the sensors and checks whether the value has reached the threshold level. If the threshold is reached, then it will send the information to the Wi-Fi to trigger the alert notifications.

2) Ultrasonic Sensors (HC-SR04): Measure the water levels and transmit the readings to the Arduino.

3) Wi-Fi module : Receives data from the Arduino and sends the alerts and data to the users via the internet using Telegram app.

4) Buzzer: Provides sound alerts when the water level reaches a higher level.

5) LED : Lights up when the water level reaches a higher level.

- Red Light: Danger.
- Blue Light: Time to take precautions.
- Green Light: Safe.

C. Software Components

The Flood Monitoring System and its Notification System are programmed, monitored, and controlled using the Arduino IDE and the Wi-Fi module.

Arduino IDE: Used for programming the microcontroller and configuring sensor data processing.

Cloud Platforms: Stores sensor data and provides a web interface for monitoring and management.

D. IoT Integration

Data collected by the Sensors is transmitted to a cloud platform via the Arduino. The platform analyses data to determine flood risk and triggers alerts based on predefined thresholds.

V. IMPLEMENTATION

A. Water Level Sensor :

This module is used to measure the distance from the surface of water and is a non-contact sensor. Functions:

- The ultrasonic sensor sends the waves to the surface of the water.
- It measures the time taken by the waves to return and collects the data.
- Calculating a distance based on how fast the waves travel.
- Finally, the sensor outputs the measured water level.

B. Arduino Microcontroller:

In the Flood Alerting System, the sensor is attached to the Arduino Microcontroller. The sensor sends the waves to the water surface, then the Arduino will compare the level to a limit. If the level reaches the limit, then the buzzer gives a sound alarm, and the LED will glow up The sensor's wireless technique plays a role in ensuring spot-on and consistent water level readings, making these elements effective components of a flood alerting system. To execute the Flood alerting system, the ultrasonic water level sensor is linked to the Arduino's input pins. Then the Arduino Microcontroller reads the data that was sent by the water level sensor and calculates the water level. There will be a particular level, and Arduino compares the calculated water level to the threshold value. If the water level reaches a particular level, then the buzzer triggers the sound alarm, which leads to the LED lights up. Additionally, the alert will be sent to the mobile by alert messages through the Wi-fi module.

The Arduino Microcontroller works as the brain of the Flood Alerting System, which will read sensor data and execute the code in real time. It also controls the functions of the Buzzer, LED, and Wi-Fi module. The Arduino monitors the functions and fixes some errors to view an output. From these steps, we get to know that the Arduino Microcontroller provides timely alerts and real-time monitoring.

C. Wi-Fi Module :

The Wi-Fi module is used to enable internet connectivity to send an alert to the mobile devices when there is a risk of flooding. Its functions include:

- Enabling Wi-Fi connections between the system components and the access point.
- Sending or receiving the data to or from a server or mobile application.
- Staying connected: Non-stop internet connection.

In the Flood Alerting System, the Wi-fi module connects to the router or access point. The Arduino shares water level data, collected from the water level sensor. When the water level reaches a higher point, the Wi-Fi module sends the alert message to the mobile app, allowing users to receive alerts and monitor remotely.

D. Buzzer :

The Buzzer is an alerting device that produces a sound when the water level reaches the maximum level. The primary function of the buzzer is to alert people in danger. The Buzzer is connected to the Arduino UNO's digital output pins. When the water level reaches the threshold level, then Arduino sends the signal to the buzzer, which leads to triggering an alert.

E. LED :

The LED [Light Emitting Diode] is a visual signal that activates lights to glow when the water level reaches a predefined level. The LED provides a signal alert in case of any chance of flood events. The LED is connected to the

Arduino microcontroller, which sends a signal to activate the light when the threshold level is met.

F. Breadboard and Jumper Wires :

In a Flood Alerting System, Breadboard and Jumper wires are used to interconnect the various components and to make a final design. By using a Breadboard, the components can be placed and blended without the need for soldering. Jumper wires will connect the components to the Arduino, which activates the flow of signals.

G. Power Supply :

The Power Supply offers power to the Flood Alerting System. The role of the power supply is to give constant power even when the power is out. The battery is attached to the Arduino Uno and other components, which provide power to the system. The battery is connected to the Arduino Uno and other components, providing power to the system. The battery can be replaced as needed.

VI. DESIGN METHODOLOGY

Floods are natural disasters causing severe impacts that lead to life loss and damage, especially for the people who suffer a lot during floods. To overcome this issue, it is important to have a real-time monitoring system to alert the people and authorities. The suggested Flood alerting system uses components such as an Arduino, Wi-Fi modules, and IoT technology to monitor the water levels and environmental conditions.

TABLE I
COMPONENT POWER CONSUMPTION

Component	Power Source
Arduino Uno	0.05 W (active)
Wi-Fi Module (ESP8266)	0.10 W (active)
Ultrasonic Sensor	0.05 W (active)
Buzzer	5V Regulated Power
LED Indicators	5V Regulated Power

The power consumption of system components is presented in Table 1. The system is connected to a power supply, which is controlled by a circuit to maintain a consistent voltage for the performance of components like the Arduino. The Arduino acts as a central processing unit, collecting and processing the data from multiple sensors.

The system wirelessly sends the data to an IoT-based cloud server using a Wi-fi module. The data is often updated and stored for real-time monitoring and later analysis. When sensor readings reach a particular point, the system triggers alerts, which will send SMS notifications to local authorities by using the wi-fi module. The areas with limited or no internet connection get alerts through cellular networks. The system architecture shows how all the components are connected to the Arduino.

The ultrasonic sensor is placed near the water bodies to monitor the water level. A combination of an IoT-based technology with a Flood detection system provides instant flood monitoring and alerting. People will receive alerts through Android applications in case of danger. Using IoT technology, we can access the data through this platform to stay informed about flood risks. In case of an emergency, the system will send notifications and alerts to the authorities to prevent flooding.

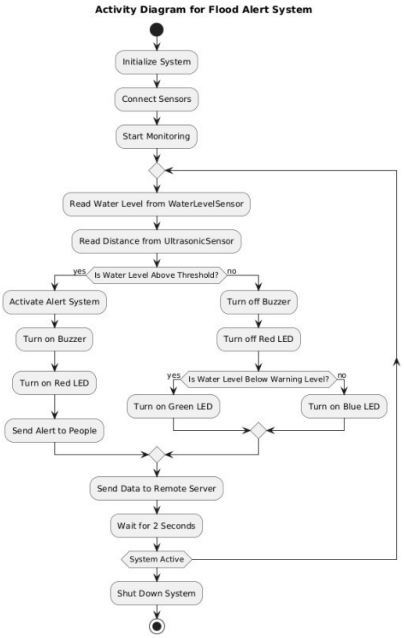

Fig. 2. Activity Diagram

As shown in Fig.2, the activity flow diagram of the flood alerting or deducting system explains the installation of the system components, followed by the connection of sensors for monitoring. After the sensors are connected, the system will start to work and monitor the level of water. The system continuously reads the level of water from the ultrasonic sensor and calculates the distance between the ground level and the water level. Then the system compares the values with a threshold value to find whether there is a flood risk or not.

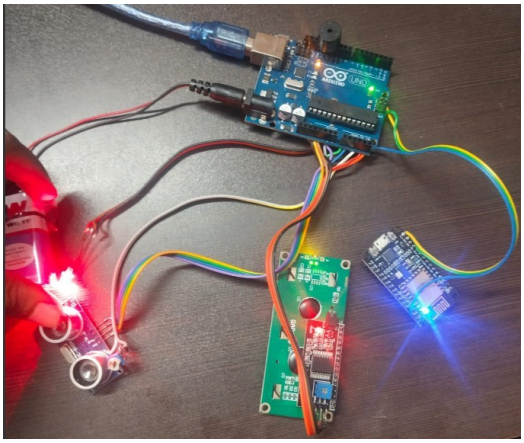

Fig. 3. Red LED indicating danger

Fig.3 shows the alert process started when the water level reaches or crosses the threshold value. The alert system starts to give alerts through the buzzer, LED, and a message. The buzzer starts to make a sound, then the red LED will glow, and an alert message will be received through the Telegram app to the nearby residents.

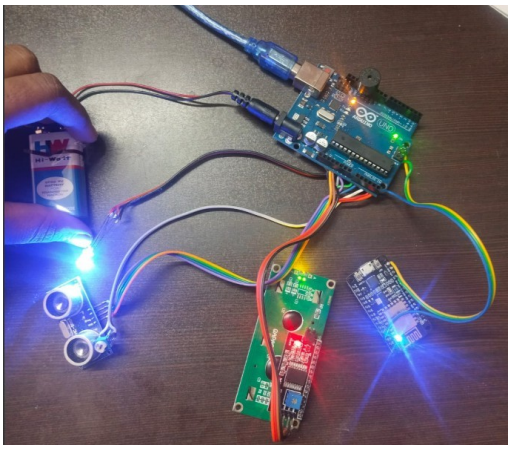

Fig. 4. Blue LED indicating warning

Fig.4 shows that if the LED glows blue, then the government or authorities have to take some precautions so that we can save people from flood risk, and it serves as a warning. Fig.5 illustrates the safe condition, where the green LED is turned on when the water level drops below the predefined risk level.

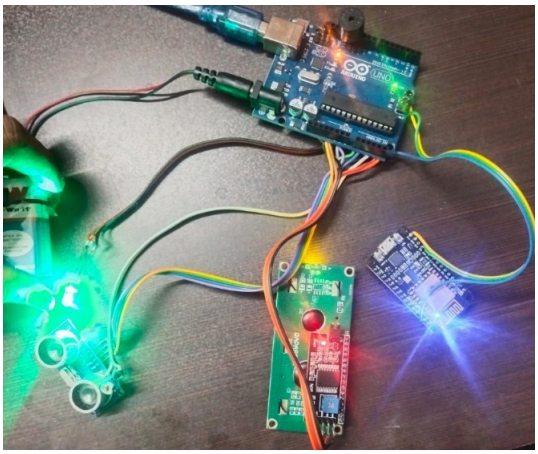

Fig. 5. Green LED indicating Safe

After each measurement, the system transfers the measured data to the Arduino UNO for real-time monitoring. Then the controller continuously monitors the water level and compares it with predefined threshold values. When the water level reaches the threshold value, the controller asks the Wi-fi module to send the alert messages to the people, and the LED indicators start to indicate according to the conditions. The system will continue the mechanism cycle and monitor the environment.

VII. DISCUSSIONS

The purpose of the flood alerting system is to warn people who are in danger. In recent years, floods have caused major consequences, especially in the place 'WAYANAD'.These incidents highlighted the need for the system and discussions on how to alert the people in flood-prone areas. Although natural disasters cannot be managed, but able to take precautions and be ready to take action. Floods have caused many losses, including human, animals, birds, etc...

The objective of this study is to develop a flood alerting system using IoT technology to provide an early warning system. Initially, various approaches for measuring the water levels were analysed through research and studies. Based on this analysis, an ultrasonic water level sensor was selected to measure the water level. The system uses IOT sensors to give a real-time data and a cloud-based server to process the data for real-time monitoring.

Environmental factors such as wind, debris and surface turbulence affect the ultrasonic sensor, which may lead to minor errors due to signal scattering. Additionally, an ultrasonic sensor is fixed in a stable position to reduce vibration-induced noise and ensure consistent distance. To overcome this issue, the system uses:

- Averaging of multiple sensor readings.
- Fixed sensor can reduce vibration.

When multiple flood zone areas trigger the alerts simultaneously, each system operates independently and sends the notifications based on a priority mechanism, ensuring risk zone areas are notified.

Telegram-based notification software was selected due to the immediate, internet-based messaging, group-based, and cost-effective communication compared to traditional SMS services. The idea is to add innovation to the system, which can enhance situational awareness.

The primary objective is to safeguard ourselves from the flood using a flood alerting system and taking some precautions beforehand.

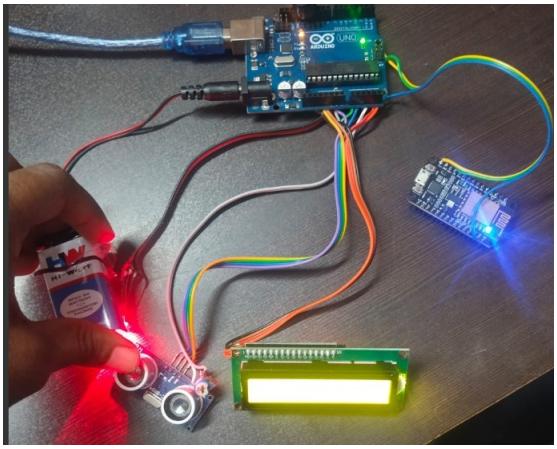

Fig. 6. Final Model

Fig.6 shows the final hardware implementation of the proposed alerting system. The proposed system was functionally tested through a real-time environment and detected changes in water level and sent alerts when the threshold values were exceeded. These results show the system is working efficiently in real-time enablin the telegram notifications.

VIII. CONCLUSION

This paper presented an IoT-based flood alerting system built for real-time water level monitoring and early warning. The system integrates the various components such as an ultrasonic sensor, Arduino microcontroller and Wi-Fi module to detect flood conditions and trigger alerts. The system combines with the IoT technology to provide advanced facilities and is suitable for deployment in flood zone areas. The system successfully detected the water level, triggered an alert when the threshold value was reached and was evaluated through real-time testing under different conditions.

Overall, the proposed flood alerting system is affordable, easy to implement and reliable for real-time monitoring. In the future, the system can be enhanced by integrating additional sensors, GPS, LORA and advanced data analytics to improve accuracy and prediction.

.

REFERENCES

[1] A.B.J, S.M., B.K. and P.R., "Machine Learning Based Advanced Flood Warning Systems", ICCCT, Chennai, India, 2025, pp.1-8, doi: 10.1109/ICCCT63501.2025.110188811.

[2] A.Kiran, P.Abhijit, V.G.Chary, B.K.Kumar, Y.V.Shivamani and R.Vivek, "Real-Time Urban Flood Forecasting and Alert System," International Conference on ASSIC, India, 2025, pp.1-5, doi: 10.1109/ASSIC64892.2025.11158584.

[3] D.M.Kalai Selvi, V.sharmila, P.Ezhumalai, M.Pranavi, P.J and P.Sadhvika, " Internet of Things-based Automated Water Level Monitoring and Alert System," ICTMIM, Kanyakumari, India, 2025, pp.1211-1217, doi:10.1109/ICTMIM65579.2025.10988382.

[4] E. A. J. A, S. J. P.Gnanaraj, K. R, and M. K. Kumar, "IoT-Powered Flood Warning and Alert System: Innovations in Flood Detection and Notification," ICSCNA, Theni, India, 2023, pp.487-490, doi: 10.1109/ICSCNA58489.2023.10370693.

[5] G.Rushingabigwi et al., "Design and Simulation of a Flood Forecasting and Alerting System," PIERS, Prague, Czech Republic, 2023, pp.905-911, doi:10.1109/PIERS59004.2023.10221434.

[6] H.K.Thakur, J.Singh, B.Verma, P.Mishra, A.Anand and M.N, "IoT-Enhanced Seaside Monitoring for Early Flood Warning Systems using Intelligent Techniques," IC3SE, Gautam Buddha Nagar, India, 2024, pp.496-500, doi:10.1109/IC3SE62002.2024.10593423.

[7] M.karthika, R.Mohan Das, E.J.D.Cuncha, A.A.Dar, G.P.Kumar and A.Kumar, "Solar Fed Flood Alert System Using Arduino," TEMSMET, India, 2023, pp. 1-7, doi: 10.1109/TEMSMET56707.2023.10150080.

[8] M.M.Myint and T.L.L.Thein, "Real-time Flood Monitoring System using Internet of Things in Urban," ICCA, Yangon, Myanmar, 2025, pp.1-4, doi: 10.1109/ICCA65395.2025.11011054.

[9] M.A.N.B.Ma'Ti,R.M.Isa,F.A.M.Kasran,S.L.M.Hassan,W.N.W.Zakaria and M.H.N.M.Ismadi,"Smart IoT-Based Flood Detection and Alert System with Real-Time Data Logging," ICSGRC, Shah Alam, Malaysia, 2025, pp.7-12, doi:10.1109.ICSGRC65918.2025.11159736.

[10] P.Nonthakarn, S.Madyod and C.Nonthakarn, "Development of a Real-Time Water Monitoring and Alert System for Sustainable Aquaculture in the Tapi River," ICEAST, Phuket, Thailand, 2025, pp.201-204, doi:10.1109/ICEAST64767.2025.11088191.

[11] R.M.D.Charaan, J.Shobana, P.Krishnamoorthy, B.A.Princy, R.J.Abinaya and K.Murugesan, "Enhancement of IoT based Flood Detection and Prevention using Arduino UNO with WiFi Module," ICACCS, Coimbatore, India, 2023, pp.2290-2294, doi: 10.1109/ICACCS57279.2023.10112673.

[12] R.Raman and S.M.U.Iqbal,"IoT-based Flood Warning System for Effective Disaster Management," ICEMPS, Thiruvananthapuram, India, 2024, pp.1-5, doi:10.1109/ICEMPS60684.2024.10559355.

[13] S.Gunanandhini, L.M, M.R, P.K.S and N.K.V, "Forecasting and Real Time Flood Monitoring System using IoT," ICACCS, Coimbatore, India, 2024, pp.1228-1223, doi:10.1108/ICACCS60874.2024.10716883.

[14] S.Naidu, A.Pandey, S.patel, K.Wagaskar, A.K.Tripathy and S.Sen, " A Flood Prediction and Management System for Kurla," ICACTA, Mumbai, India,2023, pp.1-6, doi:10.1109/ICACTA58201.2023.10392346.

[15] S.Sindhuja and V.Sravanthi, "IoT-Based Flood Monitoring and Alerting System for Smart Cities," in ISTEMS, India, 2024, pp. 1-4,doi: 10.1109/ISTEMS60181.2024.10560215

[16] V.Banupriya, I.V G, K.S, K.J, and S.R, "Flood and landslide prediction and risk assessment system using deep learning with GSM-based real-time alerts," in ICISS, India, 2025, pp.1542-1545,doi:10.1109/ICISS63372.2025.11076263.